

Li-Syue (Michael) Hsiao

v8i69g@gmail.com

SUMMARY

Digital fabrication engineer and co-founder working at the intersection of mass timber construction and robotic automation. Built and operate a KUKA robotic machining line that integrates parametric design (Rhino/Grasshopper) directly with production. Seeking digital fabrication, robotics application, or computational design (VDC) roles in North America.

EXPERIENCE

CCI Co., Ltd.

Jul 2023 – Present

Co-founder & Vice President

Chiayi, Taiwan

- Co-founded a mass timber digital fabrication and engineering firm within the Fure Jang group to localize robotic timber machining capability in Taiwan; lead a 6-person team spanning fabrication, engineering, and international procurement.
- Designed and commissioned a **KUKA robotic machining cell** for heavy timber components (Apr 2024); cut per-component machining time from 3 hours to 30 minutes (**6x**) and reduced machining error rate from **8% to 1%** on the Chiayi Railway Elevation timber structures.
- Key fabrication vendor to **Siemens Gamesa** Taiwan since 2024: deliver ~50 car-sized horizontal pre-assembly units per year for offshore wind turbine tower installation.
- Led certification program earning Taiwan Ministry of the Interior new-construction-material approvals for CLT panels (2025), including **1-hour fire-resistance** and wind-driven-rain performance ratings.
- Secured Taiwan distributorships from **Stora Enso** and K&S; built long-term supply partnerships with HASSLACHER, XLAM Dolomiti, Rotho Blaas, SIGA, and BG&E across Europe, Australia, and Japan.
- Built a **Claude API × Rhino 8** integration in **Python** (RhinoCommon): natural-language prompts generate scripts that are previewed, validated, and executed on live models — bringing AI automation into daily modeling and documentation workflows.

TAKEN Construction Material Solutions

Sep 2021 – Jan 2023

Director of Digital Fabrication

Taipei, Taiwan

- Directed digital-geometry workflows for metal building envelopes, integrating façade, fire-protection, and fabrication data into 3D models for doubly-curved surfaces — concurrently with full-time graduate study.
- Project-managed multiple envelope packages with on-site construction supervision, including the Swire Motors service center (2021) and the NTU Health Center skybridge (2022).

Fure Jang Enterprise Co., Ltd.

2018 – 2021

Assistant Engineer → Project Engineer

Taiwan

- Drove company-wide 3D/BIM transformation: replaced expert-dependent manual quantity takeoffs (1–2 hours per BOM) with **Grasshopper-automated BOM generation** at modeling time, cutting takeoff errors from **10% to 2%**; system remains in production use today (2026).
- Built Grasshopper auto-generation scripts for recurring structural assemblies, standardizing engineering output across the team.
- Concurrently supervised **four station worksites** (Neishi, Jialu, Fangshan, Longxi) on the South-Link Railway Electrification program.
- Led a **20-person team** delivering Lunping Cultural Landscape Park — Taiwan's first curved glulam structure with a cantilever exceeding **10 m** — from design development through construction (2021).
- Collaborated with **Kengo Kuma & Associates (KKAA)** on the Tsumiki Pavilion, 2018 Taichung World Flora Exposition; coordinated heavy-glulam fabrication programs in North America and imports to Taiwan.

EDUCATION

National Taiwan University of Science and Technology (Taiwan Tech)

M.S. in Architecture — Digital Fabrication, GAS Lab (architectural geometry & robotic fabrication) *Taipei, Taiwan*

- Thesis: *Digital Fabrication of the Timber Structure in Jiabei Station* — parametric integration of steel-timber hybrid connections and robotic machining (KUKA KR60: milling, slotting, drilling path tests).

TECHNICAL SKILLS

Computational Design: Rhino 3D, Grasshopper (parametric BOM & assembly generators), Python (RhinoCommon, AI-assisted), AutoCAD

Robotic Fabrication: KUKA KRL path programming, Robots plugin for Grasshopper, robotic milling/slotting/drilling

Domain: Mass timber (CLT/GLT), steel-timber hybrid connections, LOD 400 BIM coordination, construction PM

Languages: Mandarin Chinese (native), English (professional working proficiency)